

Store Retail Operations Analysis

Walmart Retail Dataset

Sales Records: 421,570 | Stores: 45 | Depts: 81
Period: 2010-02-05 ~ 2012-10-26

Key Metrics

Metric	Value
Total Sales	\$6.74B
Avg Weekly	\$15,981
Holiday Avg	\$17,036
Non-Holiday Avg	\$15,901
A-Store Avg	\$20,100
B-Store Avg	\$12,237
C-Store Avg	\$9,520
Size-Sales Corr	0.8072

Key Findings

- Store size correlates strongly with sales ($r=0.8072$). A-type stores (22 stores) average \$20,100/week, while C-type stores (6 stores) average \$9,520/week.
- Holiday weeks average \$17,036 vs \$15,901 for non-holidays -- a +7.1% lift. However, department-level impact varies from +303% to -152%.
- Department 92 leads with \$483,943,342 total sales across 45 stores. Several departments show negative sales, indicating data quality issues.
- December is the peak month, while January is the lowest. Seasonal patterns are consistent across both years.
- External factors (CPI, unemployment, fuel price, temperature) show weak direct correlation with sales at the store level.

Thinking Models

1. Decomposition

Evidence: Total sales \$6.74B broken down by store type, department, month, and holiday status. A-type stores contribute 63% of sales.

Takeaway: Break aggregates before acting.

2. Segment Divergence

Evidence: Dept sales range from \$484M (dept 92) to negative values. Holiday lift varies from +303% to -152%. Department strategy must be segment-specific.

Takeaway: Never average across highly divergent segments.

3. Proxy Inference

Evidence: CPI and unemployment proxy for consumer purchasing power. Even when sales look stable, economic indicators may signal future decline.

Takeaway: Use leading indicators, not just lagging ones.

4. Leverage Point

Evidence: Top 3 departments (92, 95, 38) generate significant revenue. Holiday-sensitive departments offer high-leverage promotional ROI.

Takeaway: Focus optimization on high-share, high-sensitivity segments.

5. Constraint vs Preference

Evidence: Store type differences reflect location strategy (constraint), not consumer preference. A-type stores are larger because of market size, not because customers prefer them.

Takeaway: Differentiate structural constraints from preference signals.